

6.2 Flows in networks

In this section we give a brief introduction to the kind of network flow theory that is now a standard proof technique in areas such as matching and connectivity. By way of example, we shall prove a classic result of this theory, the so-called *max-flow min-cut* theorem of Ford and Fulkerson. This theorem alone implies Menger's theorem without much difficulty (Exercise 3), which indicates some of the natural power lying in this approach.

Consider the task of modelling a network with one source s and one sink t , in which the amount of flow through a given link between two nodes is subject to a certain capacity of that link. Our aim is to determine the maximum net amount of flow through the network from s to t . Somehow, this will depend both on the structure of the network and on the various capacities of its connections – how exactly, is what we wish to find out.

$G = (V, E)$
 s, t, c, N
 network
 flow

Let $G = (V, E)$ be a multigraph, $s, t \in V$ two fixed vertices, and $c: \vec{E} \rightarrow \mathbb{N}$ a map; we call c a *capacity function* on G , and the quadruple $N := (G, s, t, c)$ a *network*. Note that c is defined independently for the two orientations of an edge. A function $f: \vec{E} \rightarrow \mathbb{R}$ is a *flow* in N if it satisfies the following three conditions (Fig. 6.2.1):

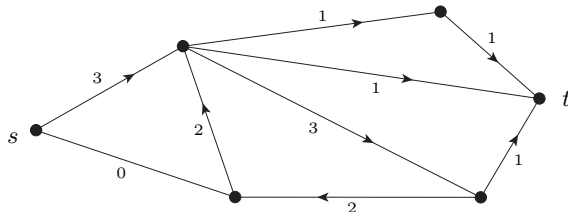


Fig. 6.2.1. A network flow in short notation: all values refer to the direction indicated (capacities are not shown)

- (F1) $f(e, x, y) = -f(e, y, x)$ for all $(e, x, y) \in \vec{E}$ with $x \neq y$;
- (F2') $f(v, V) = 0$ for all $v \in V \setminus \{s, t\}$;
- (F3) $f(\vec{e}) \leq c(\vec{e})$ for all $\vec{e} \in \vec{E}$.

integral
 f

We call f *integral* if all its values are integers.

cut in N
 capacity

Let f be a flow in N . If $S \subseteq V$ is such that $s \in S$ and $t \in \bar{S}$, we call the pair $(S, \bar{S})$ a *cut in N* , and $c(S, \bar{S})$ the *capacity* of this cut.³

Since f now has to satisfy only (F2') rather than (F2), we no longer have $f(X, \bar{X}) = 0$ for all $X \subseteq V$ (as in Proposition 6.1.1). However, the value is the same for all cuts:

³ The number $c(S, \bar{S})$ is defined in (1) of Section 6.1.

Proposition 6.2.1. *Every cut $(S, \bar{S})$ in N satisfies $f(S, \bar{S}) = f(s, V)$.*

Proof. As in the proof of Proposition 6.1.1, we have

$$\begin{aligned} f(S, \bar{S}) &= f(S, V) - f(S, S) \\ &\stackrel{(F1)}{=} f(s, V) + \sum_{v \in S \setminus \{s\}} f(v, V) - 0 \\ &\stackrel{(F2')}{=} f(s, V). \end{aligned}$$

□

The common value of $f(S, \bar{S})$ in Proposition 6.2.1 will be called the *total value* of f and denoted by $|f|$;⁴ the flow shown in Figure 6.2.1 has total value 3.

total
value $|f|$

By (F3), we have

$$|f| = f(S, \bar{S}) \leq c(S, \bar{S})$$

for every cut $(S, \bar{S})$ in N . Hence the total value of a flow in N is never larger than the smallest capacity of a cut. The following *max-flow min-cut* theorem states that this upper bound is always attained by some flow:

Theorem 6.2.2. (Ford & Fulkerson 1956)

In every network, the maximum total value of a flow equals the minimum capacity of a cut.

max-flow
min-cut
theorem

Proof. Let $N = (G, s, t, c)$ be a network, and $G = (V, E)$. We shall define a sequence $f_0, f_1, f_2, \dots$ of integral flows in N of strictly increasing total value, i.e. with

$$|f_0| < |f_1| < |f_2| < \dots$$

Clearly, the total value of an integral flow is again an integer, so in fact $|f_{n+1}| \geq |f_n| + 1$ for all n . Since all these numbers are bounded above by the capacity of any cut in N , our sequence will terminate with some flow f_n . Corresponding to this flow, we shall find a cut of capacity $c_n = |f_n|$. Since no flow can have a total value greater than c_n , and no cut can have a capacity less than $|f_n|$, this number is simultaneously the maximum and the minimum referred to in the theorem.

For f_0 , we set $f_0(\vec{e}) := 0$ for all $\vec{e} \in \vec{E}$. Having defined an integral flow f_n in N for some $n \in \mathbb{N}$, we denote by S_n the set of all vertices v such that G contains an s - v walk $x_0 e_0 \dots e_{\ell-1} x_\ell$ with

S_n

$$f_n(\vec{e}_i) < c(\vec{e}_i)$$

⁴ Thus, formally, $|f|$ may be negative. In practice, however, we can change the sign of $|f|$ simply by swapping the roles of s and t .

for all $i < \ell$; here, $\vec{e}_i := (e_i, x_i, x_{i+1})$ (and, of course, $x_0 = s$ and $x_\ell = v$).

If $t \in S_n$, let $W = x_0 e_0 \dots e_{\ell-1} x_\ell$ be the corresponding s - t walk; without loss of generality we may assume that W does not repeat any vertices. Let

$$\epsilon := \min \{ c(\vec{e}_i) - f_n(\vec{e}_i) \mid i < \ell \}.$$

Then $\epsilon > 0$, and since f_n (like c) is integral by assumption, ϵ is an integer. Let

$$f_{n+1}: \vec{e} \mapsto \begin{cases} f_n(\vec{e}) + \epsilon & \text{for } \vec{e} = \vec{e}_i, \ i = 0, \dots, \ell-1; \\ f_n(\vec{e}) - \epsilon & \text{for } \vec{e} = \vec{e}_i, \ i = 0, \dots, \ell-1; \\ f_n(\vec{e}) & \text{for } e \notin W. \end{cases}$$

Intuitively, f_{n+1} is obtained from f_n by sending additional flow of value ϵ along W from s to t (Fig. 6.2.2).

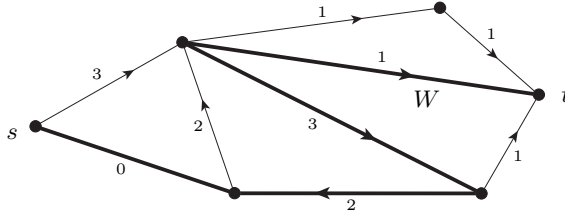


Fig. 6.2.2. An ‘augmenting path’ W with increment $\epsilon = 2$, for constant flow $f_n = 0$ and capacities $c = 3$

Clearly, f_{n+1} is again an integral flow in N . Let us compute its total value $|f_{n+1}| = f_{n+1}(s, V)$. Since W contains the vertex s only once, $\vec{e}_0$ is the only triple (e, x, y) with $x = s$ and $y \in V$ whose f -value was changed. This value, and hence that of $f_{n+1}(s, V)$ was raised. Therefore $|f_{n+1}| > |f_n|$ as desired.

If $t \notin S_n$, then $(S_n, \overline{S_n})$ is a cut in N . By (F3) for f_n , and the definition of S_n , we have

$$f_n(\vec{e}) = c(\vec{e})$$

for all $\vec{e} \in \vec{E}(S_n, \overline{S_n})$, so

$$|f_n| = f_n(S_n, \overline{S_n}) = c(S_n, \overline{S_n})$$

as desired. $\square$

Since the flow constructed in the proof of Theorem 6.2.2 is integral, we have also proved the following:

Corollary 6.2.3. *In every network (with integral capacity function) there exists an integral flow of maximum total value. $\square$*

6.3 Group-valued flows

Let $G = (V, E)$ be a multigraph. If f and g are two circulations on G with values in the same abelian group H , then $(f + g): \vec{e} \mapsto f(\vec{e}) + g(\vec{e})$ and $-f: \vec{e} \mapsto -f(\vec{e})$ are again circulations. The circulations on G with values in H thus form a group in a natural way.

$f + g$
 $-f$

An H -flow in our terminology⁵ is a circulation $f: \vec{E} \rightarrow H$ that is *nowhere zero*, one that satisfies $f(\vec{e}) \neq 0$ for all $\vec{e} \in \vec{E}$. Note that the set of H -flows on G is not closed under addition: if two H -flows add up to zero on some oriented edge $\vec{e}$, then their sum is no longer an H -flow. By Corollary 6.1.2, a graph with an H -flow cannot have a bridge.

H -flow

For finite groups H , the number of H -flows on G – and, in particular, their existence – surprisingly depends only on the order of H , not on H itself:

Theorem 6.3.1. (Tutte 1954)

For every multigraph G there exists a polynomial P such that, for any finite abelian group H , the number of H -flows on G is $P(|H| - 1)$.

Proof. Let $G = (V, E)$; we use induction on $m := |E|$. Let us assume first that all the edges of G are loops. Then, given any finite abelian group H , every map $\vec{E} \rightarrow H \setminus \{0\}$ is an H -flow on G . Since $|\vec{E}| = |E|$ when all edges are loops, there are $(|H| - 1)^m$ such maps, and $P := x^m$ is the polynomial sought.

(6.1.1)
(6.1.3)

Now assume there is an edge $e_0 = xy \in E$ that is not a loop; let $\vec{e}_0 := (e_0, x, y)$ and $E' := E \setminus \{e_0\}$. We consider the multigraphs

$e_0 = xy$
 E'

$$G_1 := G - e_0 \quad \text{and} \quad G_2 := G / e_0.$$

By the induction hypothesis, there are polynomials P_i for $i = 1, 2$ such that, for any finite abelian group H and $k := |H| - 1$, the number of H -flows on G_i is $P_i(k)$. We shall prove that the number of H -flows on G equals $P_2(k) - P_1(k)$; then $P := P_2 - P_1$ is the desired polynomial.

P_1, P_2
 k

Let H be given, and let F denote the set of all H -flows on G . Our aim is to show that

H, F

$$|F| = P_2(k) - P_1(k). \quad (1)$$

The H -flows on G_1 are precisely the restrictions to $\vec{E}'$ of those H -circulations on G that are zero on e_0 but nowhere else. Let us denote the set of these circulations on G by F_1 ; then

F_1

$$|F_1| = P_1(k).$$

⁵ To avoid cumbersome repetitions of the phrase ‘nowhere zero’ before ‘ H -flow’, we deviate slightly here from standard terminology. See the footnote in the notes.